No Petals Plucked: In-Silico Bioprospecting for the Shirui Lily

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Abstract—Range-restricted plants are appealing sources of chemical novelty, yet the destructive sampling that discovery usually requires can conflict with conservation goals. We built a provenance-tracked digital-bioprospecting workflow for metabolites relevant to the endangered Shirui lily (also called the Siroi lily), *Lilium mackliniae*, without collecting any plant material. Public records from ChEMBL and PubChem were curated for three endpoints: human 15-lipoxygenase (ALOX15) inhibition, cyclooxygenase-2 (COX-2) inhibition, and NFE2L2/antioxidant-response-element activation. On these data we compared ECFP4 tree models, a two-dimensional message-passing neural network, and a multi-conformer E(3)-equivariant graph neural network, evaluating each under scaffold-disjoint splits with strict assay-sensitivity analysis, bootstrap intervals, and applicability-domain checks. In the decisive five-fold ALOX15 experiment (202 molecules; 82 scaffolds), ECFP4-XGBoost reached an out-of-fold RMSE of 0.668 (95% scaffold-bootstrap interval 0.551–0.791), against 0.861 (0.716–1.013) for the geometric model. On the strict COX-2 data the geometric ensemble (RMSE 0.952) was statistically indistinguishable from ECFP4-Random Forest (0.940) and outperformed the two-dimensional neural model. Screening eight literature-reported *Lilium lancifolium* phenolics placed quercetin, myricetin, and kaempferol inside all three similarity domains. The results show that ligand-only three-dimensional geometry is endpoint-dependent rather than universally superior, and that conservation-oriented provenance controls are essential for responsible computational bioprospecting.

Index Terms—Computational bioprospecting, geometric deep learning, molecular graph neural network, scaffold split, natural products, conservation informatics, *Lilium mackliniae*.

1 Introduction

Natural-product discovery has traditionally begun by collecting biological material, extracting complex mixtures, and running iterative biochemical assays. For common or cultivated organisms this is usually practical. For species that are geographically restricted and already threatened, repeated sampling raises ethical, ecological, and logistical difficulties that are far harder to justify. *Lilium mackliniae* Sealy, known locally as the Shirui lily, is an accepted species of Liliaceae whose native range is centred on Manipur and Nagaland, India, and which global plant-risk assessments predict to be threatened [1]. A recent conservation review lists habitat degradation, climate pressure, fire, invasive vegetation, visitor disturbance, and direct removal among the continuing risks to its wild populations [2]. Such conditions are precisely where computational triage is valuable: reducing the number of compounds and hypotheses that require physical follow-up lowers the sampling pressure on a fragile population.

The genus *Lilium* is chemically rich, with steroidal saponins, polysaccharides, alkaloids, flavonoids, and other phenolics reported across several species [3]. Phenolic profiling of *L. lancifolium* bulbs, for instance, has identified quercetin, kaempferol, myricetin, gallic acid, and hydroxycinnamic acids [4], [5]. Reports of this kind are useful for generating transferable hypotheses, but they say nothing definite about what actually occurs in *L. mackliniae*. A responsible digital-bioprospecting study therefore has to keep three questions apart, even though they are often run together: whether a compound has been reported in a given species, whether a predictive model assigns it an activity, and whether that prediction falls inside the model's chemical applicability domain.

Molecular machine learning offers several ways to represent a compound. Extended-connectivity fingerprints encode local topological environments and remain competitive on many medicinal-chemistry data sets [12]. Random forests and gradient-boosted trees are robust in small-to-medium data regimes and use such descriptors efficiently [13], [14]. Message-passing neural networks (MPNNs) take a different route, learning atom and bond representations directly from the molecular graph [15]. Three-dimensional architectures go further still, adding interatomic distances, directions, or equivariant vector operations, as in SchNet, DimeNet, and E(n)-equivariant graph neural networks (EGNNs) [16]–[18]. Models of this last kind can, in principle, resolve stereochemical and through-space arrangements that a flat fingerprint never sees.

Whether ligand-only 3-D information pays off in practice is much less clear. Bioactivity labels are measured against proteins, cells, or whole assay systems, whereas a ligand-only model sees a single isolated conformer generated without any binding site. Flexible molecules exist as ensembles rather than one fixed structure, and the low-energy conformers a generator produces need not resemble the bioactive one [24], [25]. Evaluation adds a second complication: a random train–test division can let close analogues fall on both sides of the split and inflate the apparent ability to generalize. Separating molecules by Bemis–Murcko scaffold is a more demanding test of extrapolation to genuinely new chemotypes [19]–[21].

This paper presents a complete conservation-oriented

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computational workflow rather than assuming at the outset that a geometric model must outperform the alternatives. Its contributions are fourfold. It builds a reproducible three-endpoint benchmark from public biochemical and cellular assays. It compares ECFP4 tree models, a 2-D MPNN, and a multi-conformer E(3)-equivariant network under one shared scaffold-disjoint evaluation. It adds a strict human, biochemical, single-protein sensitivity analysis together with a five-fold scaffold cross-validation for the smallest endpoint. Finally, it applies the best validated endpoint-specific models to a provenance-labelled *Lilium*-relevant candidate set, reporting both model disagreement and chemical similarity rather than a single score. We regard the resulting negative-to-mixed outcome as informative in itself: for a conservation decision, choosing the simplest model that generalizes matters more than asserting an architectural advantage the evidence does not support.

2 Materials and Methods

2.1 Study Scope and Evidence Boundaries

The study was carried out entirely in silico. No plant, animal, human, or environmental specimen was collected, and no sensitive occurrence coordinates or traditional-knowledge records were used. *L. mackliniae* supplies the conservation use case only; the predictive models themselves were trained on public compound–assay measurements that are unrelated to any single plant species. The candidate library was assembled separately, from compounds reported in *L. lancifolium*, and every candidate row preserved the reported species, plant tissue, analytical method, citation, and a provenance tier. Throughout, predictions are read as prioritizations for future testing (against synthetic standards, cultivated material, or separately authorized sampling), not as confirmation of occurrence, efficacy, safety, or ecological function.

2.2 Endpoints and Public Data Sources

Three complementary endpoints were chosen. ALOX15 inhibition stands in for enzymatic lipid oxidation; human ALOX15 is implicated in lipid-peroxidation biology and inflammatory disease [6]. NFE2L2/NRF2 antioxidant-response-element (ARE) activation represents a cellular cytoprotective response rather than direct radical scavenging [7], [8]. COX-2 inhibition provides a clinically interpretable inflammatory endpoint [9]. Quantitative ALOX15 and COX-2 measurements were pulled through the ChEMBL REST interface from targets CHEMBL2903 and CHEMBL230; ChEMBL is a curated open database that links structures, assays, targets, and bioactivity measurements [10]. ARE records came from PubChem BioAssay AID 743219, joined to the viability counter-screen AID 743203, with PubChem providing standardized access to compound and assay records [11].

The broad benchmark kept ChEMBL activities with standard type IC_{50}, relation "=", units nM, positive numeric values, target organism *Homo sapiens*, an assay confidence of at least 8, no duplicate flag, and no serious validity warning. Regression labels were converted to pIC_{50},

$$y = -\log_{10}(\mathrm{IC}_{50}[\mathrm{M}]) = 9 - \log_{10}(\mathrm{IC}_{50}[\mathrm{nM}])\,. \qquad (1)$$

ARE compounds labelled Active or Inactive were retained once inconclusive, unspecified, and cytotoxic-confounded records had been removed. Untested compounds were never recoded as negatives.

To check whether broad aggregation was hiding assay quality, a preregistered strict subset was defined for ALOX15 and COX-2. It required ChEMBL confidence score 9; activity and assay type B (biochemical); an explicit assay organism of *Homo sapiens*; human target taxon 9606; an exact standardized IC_{50} in nM; and BAO single-protein format BAO_0000357. Replicates were aggregated first within an assay, then only across assays sharing the same BAO format, and a molecule was dropped if its compatible measurements spanned more than 2 log units. The ARE data were left untouched in this experiment, because its Tox21 reporter-and-viability design is already endpoint-specific.

2.3 Molecular Standardization and Scaffold Assignment

Structures were parsed and sanitized with RDKit, applying cleanup, parent-fragment selection, and charge neutralization in a fixed order. Molecules with fewer than three heavy atoms, or with unsupported elements, were excluded, and isomeric canonical SMILES and InChIKeys were regenerated after standardization. Duplicate regression records were collapsed to their median once the replicate-range rule had been applied; conflicting binary labels were discarded outright. Bemis–Murcko scaffolds were computed with chirality enabled. Because conventional Murcko scaffolds are empty for acyclic molecules, each acyclic structure received a connectivity-hash scaffold of its own rather than being lumped with every other acyclic compound into one artificial group.

After standardization the broad curated data held 387 ALOX15 molecules, 4,325 COX-2 molecules, and 5,419 ARE molecules. The shared scaffold-disjoint split used roughly 70% training, 15% validation, and 15% test, with every standardized structure, together with all of its conformers, confined to a single split. The strict subset contained 202 ALOX15 and 2,301 COX-2 molecules once graph-availability checks were applied. The final ALOX15 analysis used five outer scaffold folds from stratified group splitting, with an internal scaffold-disjoint validation fold drawn only from each outer development set.

2.4 Conformer Generation and Graph Construction

For each standardized molecule, explicit hydrogens were added and up to ten conformers were generated with ETKDGv3 [22], [23]. Conformers were pruned at 0.75 Å heavy-atom RMSD, optimized with MMFF94 where available and UFF otherwise, and filtered to an energy window of 10 kcal/mol above the lowest force-field energy; up to three low-energy conformers were kept.

Each atom carried categorical features (atomic number, degree, formal charge, hydrogen count, chirality, hybridization, aromaticity, ring membership, and hydrogen-bond donor and acceptor status) alongside continuous inputs for normalized atomic mass, covalent radius, van der Waals radius, and Gasteiger charge. Covalent edges recorded bond order, conjugation, ring status, stereochemistry, and an

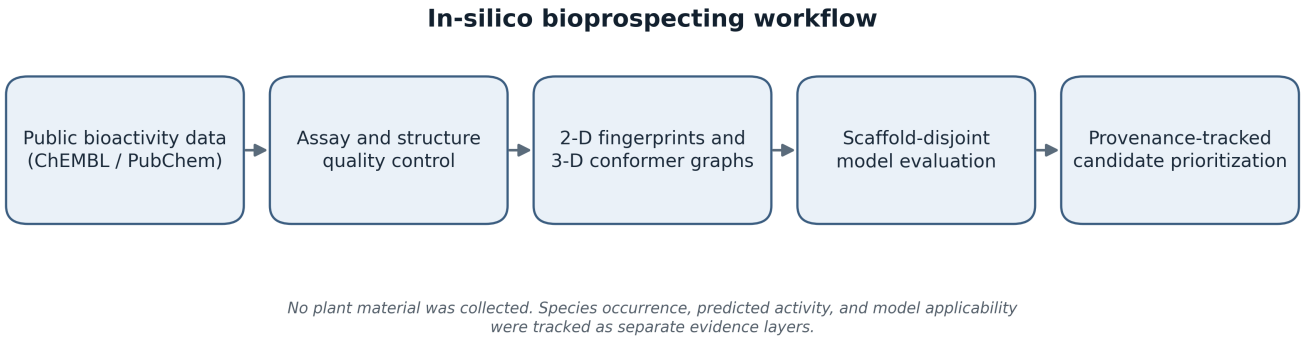


Fig. 1. Overview of the in-silico bioprospecting workflow. Public assay records were curated independently of the botanical candidate library. Molecular models were evaluated before screening, and species provenance, predicted activity, and applicability-domain evidence were tracked as separate layers. No *L. mackliniae* material was collected.

edge-type indicator. The 3-D graph added nonbonded edges within 5.0 Å, capped at 64 spatial neighbours per atom, with interatomic distance expanded into 32 Gaussian radial basis functions under a cosine cutoff.

2.5 Predictive Models

The classical representation was a 2,048-bit ECFP4 fingerprint (radius 2) [12]. Regression baselines used random forest and XGBoost; classification used balanced random forest and class-weighted XGBoost. Broad models used 600–700 estimators. The final ALOX15 cross-validation used 800 estimators with, for XGBoost, a maximum tree depth of 6, learning rate 0.04, subsampling 0.85, column subsampling 0.85, and histogram tree construction.

The 2-D neural baseline embedded atoms and passed messages along covalent edges. For layer l, a simplified update is

$$\mathbf{m}_{ij}^{(l)} = \phi_e\left(\mathbf{h}_j^{(l)}, \mathbf{a}_{ij}\right), \tag{2}$$

$$\mathbf{h}_i^{(l+1)} = \mathrm{LN}\left(\mathbf{h}_i^{(l)} + \phi_h\left(\mathbf{h}_i^{(l)}, \sum_{j\in\mathcal{N}(i)} \mathbf{m}_{ij}^{(l)}\right)\right). \tag{3}$$

The geometric model followed an EGNN-style design with scalar messages and vector-valued coordinate updates [18]. Writing $\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j$ and edge attributes $\mathbf{a}_{ij}$,

$$\mathbf{m}_{ij}^{(l)} = \phi_e\left(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \mathbf{a}_{ij}, \psi(\|\mathbf{r}_{ij}\|_2)\right), \tag{4}$$

where ψ is the radial expansion. Coordinates were updated as

$$\mathbf{x}_i^{(l+1)} = \mathbf{x}_i^{(l)} + \frac{1}{|\mathcal{N}(i)|}\sum_j \frac{\mathbf{r}_{ij}}{1+\|\mathbf{r}_{ij}\|_2}\,\phi_x\left(\mathbf{m}_{ij}^{(l)}\right). \tag{5}$$

Because the scalar messages depend only on distances and the coordinate updates are weighted relative vectors, the scalar predictions stay invariant under translation and rotation while the intermediate coordinates transform equivariantly; we verified this numerically before training. Both neural architectures used six layers, hidden dimension 256, SiLU activations, LayerNorm, residual connections, dropout of 0.08, and a readout that concatenated sum pooling, mean pooling, learned attention pooling, and the log atom count. Conformer-level predictions were averaged to the molecule level.

2.6 Training and Stabilization

Models were implemented in PyTorch 2.11 and PyTorch Geometric 2.8 [26], [27] and trained on an NVIDIA A100 80-GB GPU in Google Colab under BF16 automatic mixed precision. Irregular scatter accumulation and coordinate reductions were kept in FP32 for type compatibility and numerical stability, and atom-budget batching held each batch to roughly 16,000 atoms. Broad neural models used fused AdamW, OneCycle learning-rate scheduling, a maximum learning rate of 10^{-3}, weight decay 10^{-4}, gradient clipping at 1.0, early stopping, and five seeds (13, 37, 73, 101, 151) [28], [29].

The first strict ALOX15 runs showed substantial seed-to-seed variance, so a stabilization protocol was fixed before any further comparison. Regression targets were standardized using training data only; Smooth L1 (Huber) loss with $\beta = 1$ replaced raw MSE; the OneCycle peak learning rate was lowered to 3×10^{-4}; every molecule was given equal total loss weight regardless of how many conformers it retained; and validation RMSE was used to select checkpoints. The protocol allowed up to 120 epochs, with a minimum of 40 and patience 20. The final five-fold ALOX15 neural evaluation used three seeds per fold, averaged within each outer test fold.

2.7 Evaluation and Statistical Analysis

Regression performance was reported with RMSE, MAE, R^2, Pearson r, and Spearman ρ. ARE classification used ROC-AUC, PR-AUC, the Matthews correlation coefficient, balanced accuracy, F1, the Brier score, and expected calibration error, with PR-AUC treated as primary because the active compounds were a minority. Thresholds for the discrete classification metrics were chosen on validation data by maximizing the Matthews correlation coefficient.

Molecule-level bootstrap intervals were computed on locked test sets. The ALOX15 cross-validation instead used

TABLE 1
Broad Curated Data Sets

Endpoint	Molecules	Scaffolds	Median HA	Label summary
ALOX15	387	143	21	5.718 ± 0.869
COX-2	4,325	1,225	27	6.206 ± 1.120
ARE	5,419	2,879	14	722 active (13.3%)

scaffold-cluster bootstrap resampling, so that all compounds belonging to a sampled scaffold entered a replicate together. Paired bootstrap differences were reported as $\mathrm{RMSE}_{\mathrm{reference}} - \mathrm{RMSE}_{\mathrm{proposed}}$, so positive values favour the proposed model. Models were only ever compared on identical molecules, and the test set was never used for hyperparameter or threshold selection.

2.8 Candidate Library and Applicability Domain

Eight compounds reported in *L. lancifolium* bulb analyses were screened: quercetin, myricetin, kaempferol, caffeic acid, ferulic acid, *p*-coumaric acid, gallic acid, and protocatechualdehyde [4], [5]. No row was marked as confirmed in *L. mackliniae*. The models used for the final screen were selected on held-out evidence rather than on the candidate scores themselves: the five-fold ECFP4-XGBoost ensemble for ALOX15, ECFP4-random forest for strict COX-2, and ECFP4-XGBoost for ARE. Although the 2-D MPNN reached the highest broad ARE ensemble PR-AUC, XGBoost was kept for deployment because it was the strongest prespecified fingerprint baseline, needed no conformer generation, and reproduced more stably in operation; this choice, and its limitation, were recorded explicitly.

For every candidate, the maximum ECFP Tanimoto similarity to each endpoint's training set was computed, and a deliberately conservative heuristic threshold of 0.35 decided whether a candidate sat inside the similarity domain. EGNN predictions were retained as secondary disagreement evidence. The final exploratory score was the mean within-library percentile across ALOX15 pIC_{50}, COX-2 pIC_{50}, and ARE probability; it is a prioritization device, not a calibrated measure of therapeutic value.

3 Results

3.1 Curated Benchmark Characteristics

The broad benchmark comprised 10,131 task-labelled molecules across the three endpoints (Table 1). ALOX15 was the smallest regression set, at 387 molecules and 143 scaffolds; COX-2 held 4,325 molecules and 1,225 scaffolds; ARE held 5,419 molecules and 2,879 scaffolds, of which 722 (13.3%) were active. Median molecular size varied noticeably across endpoints, at 21 heavy atoms for ALOX15, 27 for COX-2, and 14 for ARE. The scaffold, exact-structure, label-finiteness, class-presence, checkpoint, figure, provenance, and applicability-domain validation gates all passed.

3.2 Broad Locked-Test Benchmark

The broad scaffold-disjoint benchmark gave no support to any universal geometric advantage (Fig. 3; Table 2). On ALOX15, ECFP4-random forest reached RMSE 0.820, against 0.884 for the multi-conformer EGNN ensemble and 0.993 for the 2-D MPNN; the paired EGNN improvement over random forest was -0.070, with a 95% interval of -0.192 to 0.043. On COX-2, random forest reached RMSE 0.960 while the EGNN reached 1.055, and the paired difference relative to random forest was significantly negative (-0.095, 95% interval -0.126 to -0.065). The ARE endpoint behaved differently: the 2-D MPNN ensemble reached the highest PR-AUC at 0.345 (95% interval 0.285–0.416), ahead of ECFP4-XGBoost at 0.295 and the EGNN at 0.243. Neural message passing therefore helped the cellular classification endpoint, whereas adding ligand geometry did not.

3.3 Strict Human Biochemical Sensitivity Analysis

Strict curation cut ALOX15 to 202 molecules and COX-2 to 2,301. In the initial fixed strict ALOX15 test set only 25 molecules, from eight scaffolds, survived; its mean pIC_{50} was lower and its variance narrower than in training. On this split the stabilized models produced similar RMSEs (0.519 for the 2-D MPNN, 0.539 for the EGNN, 0.553 for XGBoost, and 0.563 for random forest), but every R^2 was negative, so the small, shifted split could not settle any question of architectural superiority.

The stabilization protocol did materially reduce seed dispersion. ALOX15 EGNN seed-level RMSE tightened to 0.543 ± 0.045, against 0.559 ± 0.141 for the stabilized 2-D model. On strict COX-2, random forest reached RMSE 0.940 (95% interval 0.863–1.021), the EGNN 0.952 (0.871–1.034), XGBoost 0.971, and the 2-D MPNN 1.016. The EGNN–random forest paired interval crossed zero (-0.041 to 0.018), while the EGNN improved significantly over the 2-D MPNN by 0.065 RMSE units (0.040–0.090). Three-dimensional geometry was therefore competitive on strict COX-2, without overtaking the strongest classical baseline.

3.4 Decisive ALOX15 Five-Fold Scaffold Validation

Five-fold outer cross-validation evaluated each of the 202 strict ALOX15 molecules exactly once and separated all 82 scaffolds (Fig. 4; Table 3). ECFP4-XGBoost was strongest, with an out-of-fold RMSE of 0.668, MAE 0.491, $R^2 = 0.190$, and a scaffold-bootstrap RMSE interval of 0.551–0.791. Random forest followed at RMSE 0.693, the 2-D MPNN reached 0.821, and the multi-conformer EGNN reached 0.861 with $R^2 = -0.348$.

A paired scaffold bootstrap confirmed that this was not an artefact of one split. Relative to XGBoost, the EGNN's RMSE change was -0.190 (95% interval -0.321 to -0.073; probability of improvement 0.0002); relative to random forest it was -0.167 (-0.257 to -0.076). The EGNN-versus-2-D difference was smaller and inconclusive (-0.040, -0.095 to 0.011), which locates the principal gap between the learned neural representations and the fingerprint/tree regime rather than in any penalty unique to equivariance.

3.5 Ablations, Calibration, and Applicability

Single-seed ablations are read here as mechanistic diagnostics, not definitive rankings. On broad ALOX15, distance-informed message passing (RMSE 0.878) outperformed a

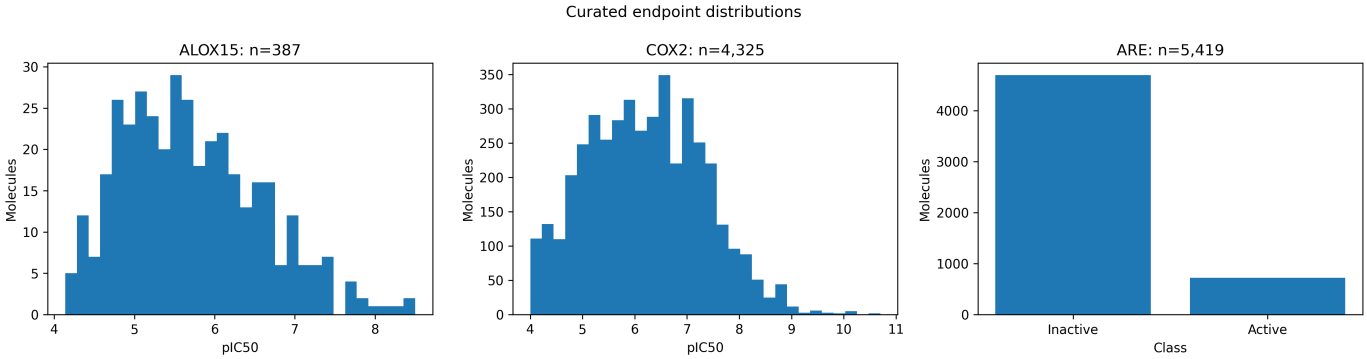


Fig. 2. Broad curated endpoint distributions after standardization and replicate resolution. ALOX15 and COX-2 labels are pIC_{50}; ARE is a binary Tox21 reporter endpoint after viability filtering.

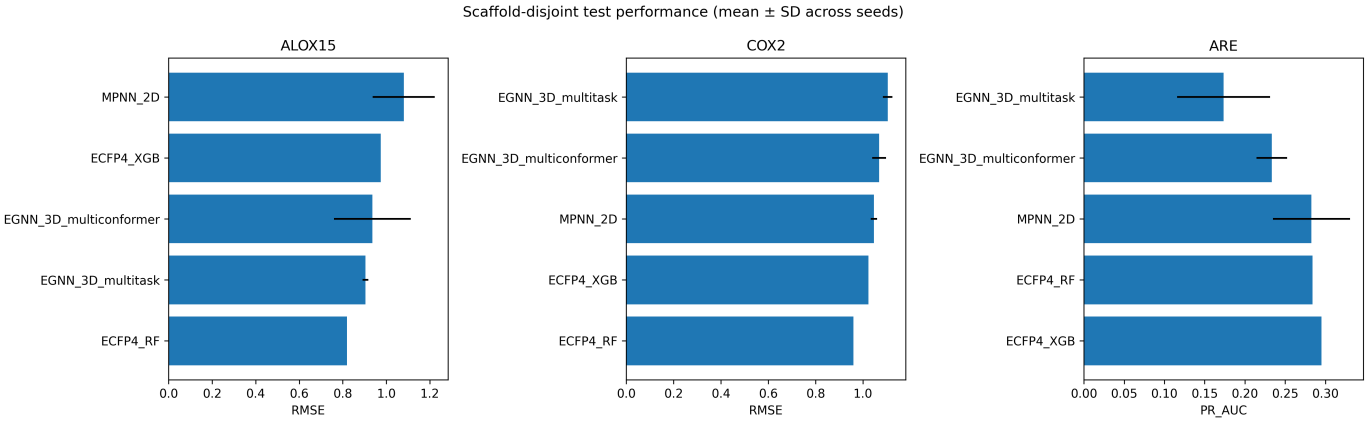


Fig. 3. Broad scaffold-disjoint test performance. Neural bars show mean performance across seeds with standard-deviation error bars. Lower RMSE is better for ALOX15 and COX-2; higher PR-AUC is better for ARE.

TABLE 2
Broad Locked-Test Performance. Intervals Are Molecule-Level Bootstrap 95% Intervals for Ensemble Predictions.

Endpoint	Model	Primary metric	Estimate	95% interval	Secondary metric
ALOX15	ECFP4-RF	RMSE	0.820	0.645–1.017	–
ALOX15	ECFP4-XGB	RMSE	0.975	0.819–1.141	–
ALOX15	2-D MPNN ensemble	RMSE	0.993	0.765–1.231	–
ALOX15	3-D EGNN ensemble	RMSE	0.884	0.769–1.006	–
COX-2	ECFP4-RF	RMSE	0.960	0.909–1.015	–
COX-2	ECFP4-XGB	RMSE	1.023	0.973–1.077	–
COX-2	2-D MPNN ensemble	RMSE	1.025	0.971–1.085	–
COX-2	3-D EGNN ensemble	RMSE	1.055	1.006–1.111	–
ARE	ECFP4-RF	PR-AUC	0.284	0.236–0.341	ROC-AUC 0.683
ARE	ECFP4-XGB	PR-AUC	0.295	0.244–0.356	ROC-AUC 0.698
ARE	2-D MPNN ensemble	PR-AUC	0.345	0.285–0.416	–
ARE	3-D EGNN ensemble	PR-AUC	0.243	0.200–0.304	–

covalent-only EGNN (1.097), whereas collapsing the conformer ensemble to a single conformer degraded RMSE to 1.521. On COX-2, the distance-only, covalent-only, and single-conformer variants all clustered near 1.05 RMSE. On ARE, the covalent-only variant scored a higher PR-AUC than the distance-only variant, again indicating endpoint dependence, and mean-only pooling was not reliably beneficial.

The ARE curves (Fig. 5) showed modest discrimination and imperfect calibration across every model class. For several regressions, similarity was a more interpretable applicability signal than neural ensemble variance: in the broad ALOX15 analysis, absolute error fell as maximum training-set similarity rose. Because the uncertainty correlations were inconsistent from endpoint to endpoint, candidate reporting kept both similarity and model disagreement instead of presenting ensemble variance as a complete uncertainty estimate.

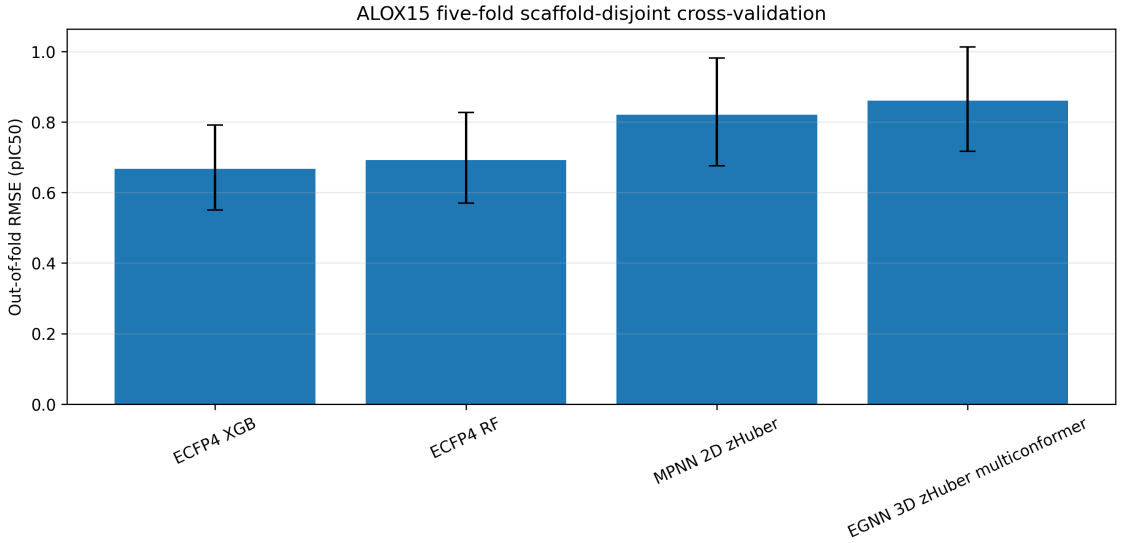


Fig. 4. ALOX15 five-fold scaffold-disjoint out-of-fold RMSE with scaffold-cluster bootstrap intervals. Each of 202 molecules and 82 scaffolds was assigned to exactly one outer test fold.

TABLE 3
ALOX15 Five-Fold Scaffold Cross-Validation

Model	RMSE	95% interval	R^2
ECFP4-XGB	**0.668**	0.551–0.791	0.190
ECFP4-RF	0.693	0.570–0.827	0.128
2-D MPNN	0.821	0.676–0.981	−0.226
3-D EGNN	0.861	0.716–1.013	−0.348

3.6 Provenance-Tracked Candidate Prioritization

Quercetin ranked first in the exploratory three-endpoint consensus, ahead of myricetin and kaempferol (Table 4), and these three flavonoids were the only candidates that fell inside the predefined similarity domain for all endpoints. Quercetin's predictions were ALOX15 pIC_{50} 5.332, COX-2 pIC_{50} 5.797, and ARE probability 0.891, with myricetin and kaempferol close behind. Caffeic acid received the highest ALOX15 prediction of all (5.555) yet fell outside the ALOX15 similarity domain, which shows why a raw activity score should not set priority on its own.

All eight candidates were reported from *L. lancifolium* bulbs, not confirmed from *L. mackliniae*. Their consensus ranks therefore identify compounds worth following up with synthetic standards or cultivated material; they are not evidence of Shirui lily chemistry. The predicted pIC_{50} values are themselves fairly modest, which further supports reading the list as exploratory rather than as a set of high-potency inhibitors.

4 Discussion

The central finding is deliberately more restrained than the original architectural hypothesis. Under rigorous, repeated, scaffold-disjoint ALOX15 evaluation, ECFP4-XGBoost and random forest generalized better than both neural graph models. The geometric model passed its invariance tests, trained stably, and generated conformers without incident, so its weaker ALOX15 performance cannot be attributed to a simple implementation failure. Instead, it shows that a representation with greater physical expressivity is not automatically better aligned with noisy, ligand-only bioactivity labels.

Several mechanisms could explain the fingerprint advantage. First, the strict ALOX15 data were small, at 202 molecules across 82 scaffolds, and tree models can exploit sparse substructure indicators at relatively low variance, whereas a six-layer neural encoder has far more parameters to estimate. Second, IC_{50} depends on protein state, substrate concentration, assay conditions, and inhibition mechanism, none of which the model ever saw. Third, ETKDG/MMFF conformers are plausible isolated geometries rather than experimentally resolved ligand–protein poses. Fourth, flexible molecules spread their probability over conformational ensembles, and uniformly averaging up to three conformers cannot guarantee that the relevant state is among them. Fifth, ECFP4 may capture medicinal-chemistry motifs that recur across assay series even when the full Bemis–Murcko scaffolds differ.

The COX-2 result rules out the stronger claim that geometry is uninformative. On 408 strict locked-test molecules the EGNN was statistically indistinguishable from random forest and significantly better than the 2-D MPNN. This endpoint-dependent pattern fits the broader observation that the usefulness of a molecular representation depends on target, data scale, assay definition, and split regime. Geometry likely offers the most benefit when 3-D pharmacophore constraints are well represented in the data, when generated conformers approximate the bound arrangement, or when explicit protein environments are available. A natural next step is a protein–ligand model built on experimentally determined or carefully docked complexes and compared under the same scaffold and target-family controls.

The study also shows how evaluation design can reverse

TABLE 4
Final Candidate Prioritization. Values Are Selected-Model Ensemble Means. "In All Domains" Requires Maximum ECFP Tanimoto Similarity ≥ 0.35 for ALOX15, COX-2, and ARE.

Rank	Compound	ALOX15 pIC_{50}	COX-2 pIC_{50}	ARE probability	Consensus	In all domains
1	Quercetin	5.332	5.797	0.891	0.833	Yes
2	Myricetin	5.296	5.790	0.877	0.667	Yes
3	Kaempferol	5.153	5.885	0.851	0.625	Yes
4	Caffeic acid	5.555	5.297	0.501	0.583	No
5	Ferulic acid	5.467	5.488	0.368	0.542	No
6	*p*-Coumaric acid	5.419	5.402	0.375	0.500	No
7	Gallic acid	5.318	5.162	0.751	0.458	No
8	Protocatechualdehyde	5.248	4.965	0.575	0.292	No

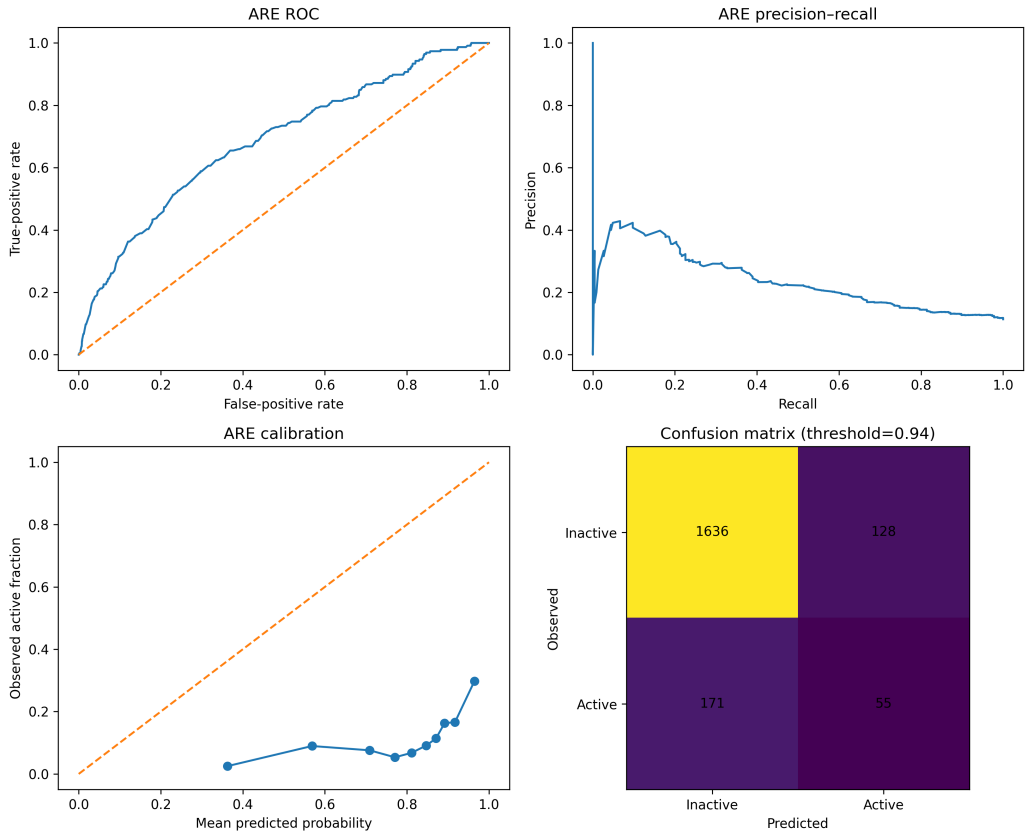


Fig. 5. Broad ARE receiver-operating-characteristic, precision–recall, and calibration curves. The endpoint is imbalanced (13.3% active), making PR-AUC and calibration essential complements to ROC-AUC.

an apparent result. The small fixed ALOX15 strict split hinted at rough parity among the models, but its 25 test molecules came from only eight scaffolds and carried a shifted label distribution. Turning the full strict set into out-of-fold evidence through five-fold scaffold cross-validation revealed a clear fingerprint advantage that the favourable fixed split had hidden; reporting only that split would have been misleading. Repeated or grouped validation matters especially in natural-product work, where data sets are small and the candidate chemistry can sit far from conventional drug-like training compounds.

From a conservation standpoint, the contribution is not a claim that computation can replace experimental biology. It is a way to order hypotheses before anything is collected. The workflow preserved botanical provenance, refused species-specific overclaims, withheld sensitive location data, and required applicability-domain reporting. Quercetin, myricetin, and kaempferol emerged as the most defensible transferable candidates precisely because literature provenance and chemical similarity supported their predictions across all three endpoints. Their ranking gives a rationale for testing commercially available standards or cultivated *Lilium* material first, which in turn reduces the pressure to sample a threatened wild population.

Several limitations apply. The candidate library holds only eight phenolics from *L. lancifolium*; it is neither a genus-level metabolome nor evidence for *L. mackliniae*. Public assay data stay heterogeneous even after strict filtering, and the broad ARE endpoint is cellular while ALOX15 and COX-2 are biochemical. No docking, protein sequence,

protein structure, pharmacokinetic, toxicity, or selectivity information was included. The similarity threshold of 0.35 is a transparent heuristic, not a universal applicability boundary. Bootstrap intervals quantify sampling variation under the observed data structure but do not capture every source of assay or model uncertainty. Finally, there was no wet-lab confirmation.

Future work should prioritize non-destructive or minimal-sample metabolomics conducted under appropriate conservation permissions and local governance, followed by structural confirmation against authentic standards. A larger candidate library ought to include steroidal saponins and other characteristic *Lilium* metabolites, with stereochemistry and tissue provenance preserved. On the computational side, the benchmark can be extended with protein-informed equivariant models, conformer reweighting, temporal splits, assay-context embeddings, and calibrated conformal prediction. Whatever is added should keep the negative-result discipline used here: new complexity has to earn its place through out-of-family evidence, not through training performance.

5 Conclusion

An A100-accelerated pipeline was developed for conservation-oriented digital bioprospecting across the ALOX15, COX-2, and NRF2/ARE endpoints. The decisive ALOX15 scaffold cross-validation showed ECFP4-XGBoost outperforming a multi-conformer E(3)-equivariant network, while the strict COX-2 results showed the same geometric model remaining competitive and beating a 2-D neural baseline. Ligand-only three-dimensional learning was therefore endpoint-dependent, not universally superior. Provenance and applicability controls singled out quercetin, myricetin, and kaempferol as the most defensible transferable *Lilium*-relevant candidates, while stopping short of any claim that they occur in *L. mackliniae*. The study offers a reproducible framework for narrowing experimental search spaces around conservation-sensitive species, without presenting computation as a substitute for chemical identification or biological validation.

Data and Code Availability

The complete executable Google Colab notebook, the immutable public-database snapshots used in the experiments, the curated endpoint tables, exclusion audits, scaffold-split manifests, out-of-fold predictions, model metrics, figure files, and a SHA-256 handoff manifest are provided as Supplementary Data with this submission. ChEMBL and PubChem records remain subject to their respective database terms. The candidate table carries source citations and provenance fields for every compound.

Ethics and Conservation Statement

This work used public molecular and assay data only. No human participants, animals, plant specimens, field locations, or traditional-knowledge records were involved, so institutional ethics approval did not apply. The predictions reported here must not be used to justify collecting wild *L. mackliniae* without the relevant scientific, legal, and community permissions.

Conflicts of Interest

The author declares no competing interests.

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